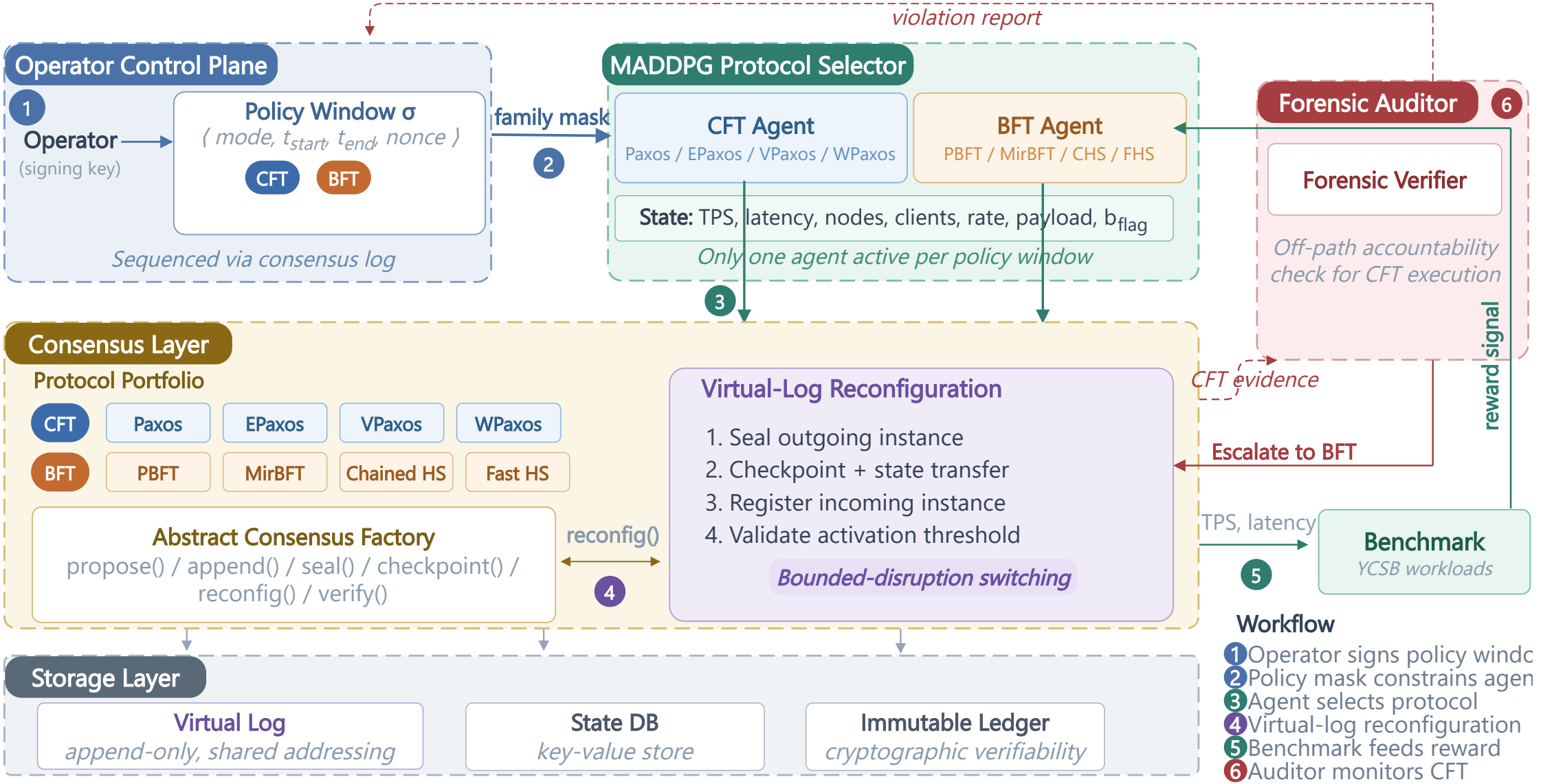

violation report
Operator Control Plane
1
Operator
(signing key)
Policy Window σ
⟨ mode, t_{start}, t_{end}, nonce ⟩
CFT
BFT
Sequenced via consensus log
family mask
2
MADDPG Protocol Selector
CFT Agent
Paxos / EPaxos / VPaxos / WPaxos
BFT Agent
PBFT / MirBFT / CHS / FHS
State: TPS, latency, nodes, clients, rate, payload, b_{flag}
Only one agent active per policy window
3
Forensic Auditor
6
Forensic Verifier
Off-path accountability check for CFT execution
reward signal
CFT evidence
Escalate to BFT
Consensus Layer
Protocol Portfolio
CFT
Paxos
EPaxos
VPaxos
WPaxos
BFT
PBFT
MirBFT
Chained HS
Fast HS
Abstract Consensus Factory
propose() / append() / seal() / checkpoint() / reconfig() / verify()
reconfig()
4
Virtual-Log Reconfiguration
1. Seal outgoing instance
2. Checkpoint + state transfer
3. Register incoming instance
4. Validate activation threshold
Bounded-disruption switching
TPS, latency
5
Benchmark
YCSB workloads
Storage Layer
Virtual Log
append-only, shared addressing
State DB
key-value store
Immutable Ledger
cryptographic verifiability
Workflow
1 Operator signs policy windo
2 Policy mask constrains agen
3 Agent selects protocol
4 Virtual-log reconfiguration
5 Benchmark feeds reward
6 Auditor monitors CFT